

Virtualization – Fundamentals and General Idea

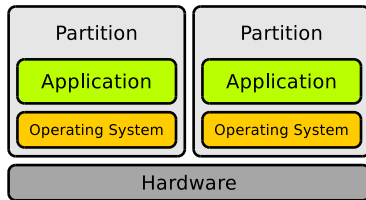
- By using **virtualization**, the resources of a computer system can be split and used by multiple independent operating system instances
- Several fundamentally different approaches and technologies exist to implement virtualization
- Each **virtual machine** (VM)...
 - behaves like any other computer, with own components
 - runs inside an isolated environment on a physical machine
- Inside a VM, an operating system with applications can be installed, exactly like on a physical computer
 - The applications do not notice that they are located inside a VM
- Requests from the operating system instances are transparently intercepted by the virtualization software and converted for the existing physical or emulated hardware
 - The VM itself does not become aware of the virtualization layer between itself and the physical hardware

- Creasy RJ. **The origin of the VM/370 time-sharing system.** IBM Journal of Research and Development (1981), No. 5, P.483–490

- Prof. Dr. Christian Baun – 10th Slide Set Operating Systems – Frankfurt University of Applied Sciences – WS2627

Partitioning

- If partitioning is used, the total amount of resources can be split to create subsystems of a computer system
 - Each subsystem may contain an executable operating system instance
 - Each subsystem can be used like an independent computer system
- The resources (CPU, main memory, storage. . .) are managed by the **firmware** of the computer and assigned to the VMs
- Partitioning is used, e.g. in IBM mainframes (zSeries) and midrange systems (pSeries) with Power5/6/7 CPUs
 - Resource allocation is possible during operation without having to restart
 - On a modern mainframe computer several hundred to thousands of Linux instances can operate simultaneously
- Modern CPUs only support the partitioning of the CPU itself and not of the entire system (Intel Vanderpool, AMD Pacifica)
 - **Partitioning is not used for desktop environments**



- February 2011: *Watson* wins the Quiz *Jeopardy Challenge* in the U.S.
 - Watson is a cluster of 90 IBM Power 750 servers with 2,880 Power7 CPU cores (each with 8 cores per CPU) and 16 TB RAM



Image (Interns demonstrating Watson capabilities in Jeopardy! exhibition match): Rosemaryetoufee. Wikimedia (CC-BY-SA-4.0)

Partitioning Example – Watson (2/2)

- Partitions can be created at each one of the 90 nodes
 - Each partition may contain an AIX, Linux or IBM i (formerly OS/400)
 - The partitions are independent installations
 - Each partition can contain a different operating system
- On each node runs a *POWER Hypervisor*
 - It controls the hardware access
- Since Power6, running partitions can be relocated without interruption from one physical server to another one ($\implies$ Live Partition Mobility)
- Partitions can share main memory ($\implies$ Active Memory Sharing)
 - Active Memory Expansion is able to compress storage pages
 - Depending on the application, compression is faster compared with relocating or swapping

-
- The diagram illustrates the layers of a virtual machine architecture. It consists of five stacked components:
- Application:** A large green box on the left, representing the user application.
 - Emulated Guest System:** A grey box on the right containing:
 - Application:** A green box representing the application running inside the guest.
 - Guest OS:** A blue box representing the operating system running inside the guest.
 - Emulator:** A green box located below the Emulated Guest System, representing the software that emulates the guest hardware.
 - Host OS:** An orange box located below the Emulator, representing the operating system on the physical hardware.
 - Hardware:** A grey box at the bottom, representing the physical hardware.

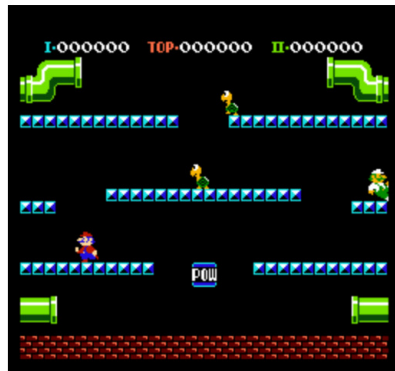
[illegible]

Example of a current Emulator - JSNES

Ben Firshman

JSNES

- JSNES emulates the Nintendo Entertainment System (NES)
- The emulator is implemented in JavaScript and executes in the browser
- <https://jsnes.org>
- github.com/bfirsh/jsnes
- Free Software (GPLv3)



Mario Bros. ▾
pause restart enable sound zoom out

Running: 44.40 FPS

Latest Development: Browser emulates PC – jslinux

<https://www.wired.com/2011/05/yes-virginia-that-is-linux-running-on-javascript/>

Date: May 18th 2011

Author: Scott Gilbertson

JavaScript never seems to get any respect. It's not a real programming language, detractors complain, it's just some script language that runs in the web browser. We're not sure what makes JavaScript less „real“ to some, but thanks to today's web browsers, JavaScript has become a very powerful language. Powerful enough to run Linux in your web browser. French developer Fabrice Bellard has built a **JavaScript-based x86 PC emulator capable of running Linux inside a web browser**.

If you'd like to try it out, point Firefox 4 or Chrome 11 to the demo page. Keep in mind that this is just Linux, no X Window or other graphical interface, just the command line, a small C compiler and QEmacs, Bellard's emacs clone. Still, it's really Linux, really running in your web browser, really using JavaScript to emulate hardware.

```
TCP reno registered
checking if image is initramfs...it isn't (bad gzip magic numbers); looks like a
n initrd
Freeing initrd memory: 2048k freed
Total HugeTLB memory allocated, 0
io scheduler noop registered
io scheduler anticipatory registered
io scheduler deadline registered
io scheduler cfq registered (default)
Real Time Clock Driver v1.12ac
JS clipboard: I/O at 0x03c0
Serial: 8250/16550 driver $Revision: 1.90 $ 4 ports, IRQ sharing disabled
serial8250: ttyS0 at I/O 0x3f8 (irq = 4) is a 16450
RAMDISK driver initialized: 16 RAM disks of 4096K size 1024 blocksize
loop: loaded (max 8 devices)
TCP cubic registered
NET: Registered protocol family 1
NET: Registered protocol family 17
Using IPI Shortcut mode
Time: pit clocksource has been installed.
RAMDISK: ext2 filesystem found at block 0
RAMDISK: Loading 2048KiB [1 disk] into ram disk... done.
EXT2-fs warning: maximal mount count reached, running e2fsck is recommended
VFS: Mounted root (ext2 filesystem).
Freeing unused kernel memory: 124k freed
Booted in 8.961 s
Welcome to JS/Linux
# uname -a
Linux (none) 2.6.20 #2 Mon Aug 8 23:51:02 CEST 2011 i586 GNU/Linux
# #
© 2011 Fabrice Bellard - http://bellard.org/jslinux/
```

Image Source: <http://bellard.org/jslinux/>

← → ↻ bellard.org/jslinux/

JSLinux

Run Linux or other Operating Systems in your browser!

The following emulated systems are available:

CPU	OS	User Interface	VFSync access	Startup Link	TEMU Config	Comment
x86	Alpine Linux 3.12.0	Console	Yes	click here	url	
x86	Alpine Linux 3.12.0	X Window	Yes	click here	url	Right mouse button for the menu.
x86	Windows 2000	Graphical	No	click here	url	Disclaimer
x86	FreeDOS	VGA Text	No	click here	url	
risecv64	Buildroot (Linux)	Console	Yes	click here	url	
risecv64	Buildroot (Linux)	X Window	Yes	click here	url	Right mouse button for the menu.
risecv64	Fedora 29 (Linux)	Console	Yes	click here	url	Warning: longer boot time.
risecv64	Fedora 29 (Linux)	X Window	Yes	click here	url	Warning: longer boot time. Right mouse button for the menu.

© 2011-2020 Fabrice Bellard - [News](#) - [VM list](#) - [FAQ](#) - [Technical news](#)

← → ↻ bellard.org/jslinux/vm.html?url=alpine-x86.cfg&mem=192

```

Loading...

Welcome to JS/Linux (1586)

Use 'vlogin username' to connect to your account.
You can create a new account at https://vfsync.org/signup .
Use 'export file filename' to export a file to your computer.
Imported files are written to the home directory.

localhost:~# uname -a
Linux localhost 4.12.0-rc6-g48ec1f0-dirty #21 Fri Aug 4 21:02:28 CEST 2017 i586
Linux
localhost:~# █

```

- Since 2011, the author of JSLinux has added a lot of new features

Image top right: FreeDOS 1.2 (x86)

Image bottom left: Alpine Linux 3.12.0 (x86)

Image bottom right: Windows 2000 (x86)

← → ↻ bellard.org/jslinux/vm.html?url=freedos.cfg&mem=64&graphic=1&w=720&h=400

```

Type HELP to get support on commands and navigation.

Welcome to the FreeDOS 1.2 operating system (http://www.freedos.org)

Use KEYB to set the keyboard mapping (e.g. KEYB FR for a French keyboard)

C:\>mem

Memory Type      Total      Used      Free
-----
Conventional      639K       33K       606K
Upper              0K         0K         0K
Reserved          385K       385K         0K
Extended (XMS)    64,512K     320K     64,104K
-----
Total memory      65,536K     746K     64,790K

Total under 1 MB   639K       33K       606K

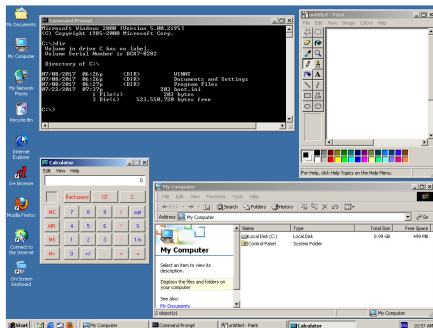
Total Expanded (EMS)      8,680K (8,096,512 bytes)
Free Expanded (EMS)      8,192K (8,388,608 bytes)

Largest executable program size      696K (620,000 bytes)
FreeDOS is resident in the high memory area.

C:\>

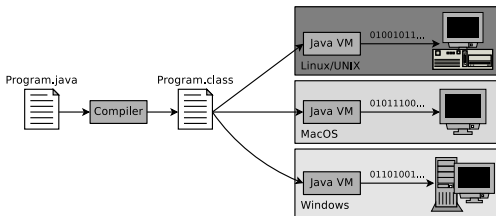
```

← → ↻ bellard.org/jslinux/vm.html?url=win2k.cfg&mem=192&graphic=1&w=1024&h=768



Application Virtualization

- Applications are executed inside a virtual environment, which uses local resources and provides all the components the application needs
 - The VM is between the executed application and the operating system
- Popular example: Java Virtual Machine (JVM)
 - The JVM is the part of the Java Runtime Environment (JRE), which executes the Java bytecode
 - The JVM is for Java programs the interface to the computer system and its operating system



- The compiler javac compiles source code into architecture-independent .class files, which contain bytecode, that can be executed in the Java VM
- The program javac launches a Java application inside a Java VM

- Advantage: Platform independence
- Drawback: Less performance, compared to native program execution

VMware ThinApp

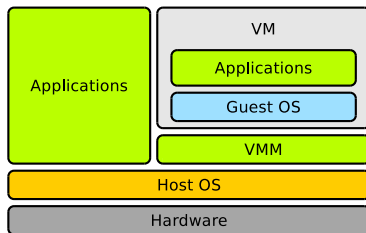
<http://www.vmware.com/products/thinapp/>

- Further example of application virtualization: VMware ThinApp
 - Until 2008, the software was named Thinstall
- Packs Windows applications into a single .exe file
- The application becomes portable and can be used without local installation
 - Applications can, e.g. be executed from an USB flash memory drive
- No entries are inserted into the Windows registry and no environment variables or DLL files are created on the system
- User preferences and created documents are stored inside a separate sandbox
- Drawback: The software only supports Microsoft Windows

The boundaries between Application Virtualization and Operating System-level Virtualization / Containers (see slide 26) are not sharp

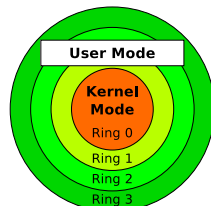
Full Virtualization (1/3)

- Full virtualization software solutions offer each VM a complete virtual PC environment, including an own BIOS
 - Each guest operating system gets its own VM with virtual resources (e.g. CPU, main memory, storage devices, network adapters)
- A **Virtual Machine Monitor** (VMM) is used
 - The VMM is also called **Type-2 hypervisor**
 - The VMM runs *hosted* as an application in the host operating system
 - The VMM distributes hardware resources to VMs
- Some hardware components are emulated, because they are not designed for the concurrent access by multiple operating systems
 - Example: Network adapters
 - The emulation of popular hardware avoids driver issues

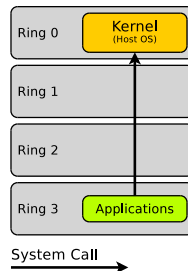


Virtualization Basics of the x86 Architecture

- x86-compatible CPUs implement 4 privilege levels
 - Objective: Improve stability and security
 - Each process is assigned to a ring permanently
- Implementation: The register CPL (Current Privilege Level) stores the current privilege level
- Source: <http://css.csail.mit.edu/6.858/2012/readings/i386.pdf>
- In ring 0 (= **kernel mode**) runs the kernel
 - Processes here have full hardware access
- In ring 3 (= **user mode**) run the applications
 - Processes are in Protected Mode ($\Rightarrow$ slide set 5)
- Unmodified modern operating systems only use 2 privilege levels (rings) $\Rightarrow$ Rings 0 and 3
- If a user-mode process must carry out a higher privileged task (e.g. accessing hardware), it can notify the kernel via a **system call** ($\Rightarrow$ slide set 7)
 - The user-mode process generates an exception, which is intercepted in ring 0 and handled there

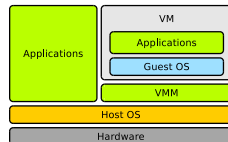
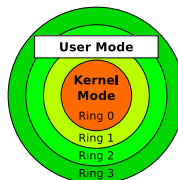
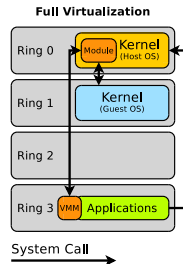


Without Virtualization



Full Virtualization (2/3)

- Full virtualization makes use of the fact, that x86 systems typically use only 2 privilege levels
 - The VMM runs together with the applications in ring 3
 - VMs are in the less privileged ring 1
- The VMM contains for every exception a treatment, which catches, interprets and executes privileged operations of guest operating systems
- VMs can only access the hardware via the VMM
 - This ensures controlled access to shared system resources



Full Virtualization (3/3)

- Advantages:

- Few modifications in the host and guest operating systems are required
- Access to the main resources is just forwarded (passed through)
⇒ guest operating systems run almost with native performance
- Each guest operating system has its own kernel
⇒ high degree of flexibility

- Drawbacks:

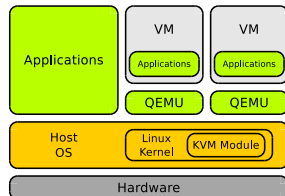
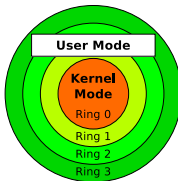
- Switching from one ring to another one requires a process/context switch
⇒ each process/context switch consumes CPU time
- If an application in the guest operating system requests the execution of a privileged instruction, the VMM provides a replacement function, which commands the execution via the kernel API of the host operating system
⇒ speed losses

Full Virtualization Examples: Some virtualization solutions, which implement the VMM concept

- VirtualBox
- VMware Server, VMware Workstation and VMware Fusion
- Parallels Desktop and Parallels Workstation
- Kernel-based Virtual Machine (KVM) ⇒ The architecture of KVM is different. . . (see next slide)

Kernel-based Virtual Machine (KVM)

- KVM is integrated as a module directly in the Linux kernel
 - KVM core module: `kvm.ko`
 - Hardware-specific modules: `kvm-intel.ko` and `kvm-amd.ko`
- After loading the modules, the kernel itself operates as a hypervisor
- KVM can only operate with CPUs, which implement hardware-assisted virtualization
- Besides the kernel modules, KVM contains the emulator QEMU
 - KVM does not provide virtual hardware. This is provided by QEMU
 - CPU virtualization provides the CPU (Intel VT or AMD-V)
 - Main memory and storage are virtualized by KVM
 - I/O is virtualized by a dedicated QEMU process per guest



Paravirtualization (1/2)

- No hardware is virtualized or emulated
 - Does not provide an emulated hardware layer to the guest operating systems, but only an application interface
- Guest operating systems use an abstract management layer (⇒ **hypervisor**) to access the physical resources
 - Hypervisor is a **meta operating system**, which is reduced to a minimum
 - The hypervisor distributes hardware resources among the guest systems, the same way, an operating system would distribute hardware resources among running processes
 - The hypervisor is a **Type-1 hypervisor** and runs *bare metal*
 - A meta operation system allows the independent operation of different applications and operating systems on a single CPU
- The hypervisor runs in the privileged ring 0
 - The guest operating systems are relocated to the less privileged ring 1
 - The hypervisor must include all required device drivers or as alternative, a guest operating system instance that includes the required device drivers is mandatory

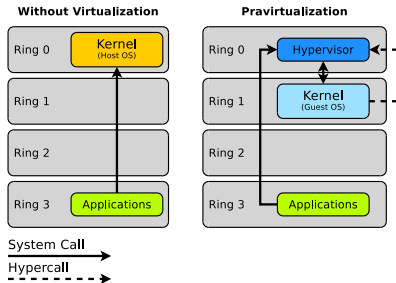
Paravirtualization (2/2)

- Guest operating system kernels cannot execute privileged instructions
 - Solution: The hypervisor provides **hypercalls**

Hypercalls are similar to system calls

- The interrupt numbers are different
- If an application requests the execution of a system call, a replacement function in the hypervisor is called
- The hypervisor orders the execution of the system call via the kernel API of the operating system

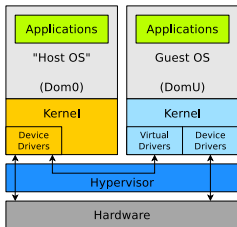
- Guest operating kernels need to be modified in a way that any system call for hardware access is replaced by the corresponding hypercall
- Catching and verifying system calls by the hypervisor causes just little performance loss



Paravirtualization Examples: Some virtualization solutions, which implement the paravirtualization concept

Examples: Xen, Citrix Xenserver, Virtual Iron, VMware ESX Server

Xen



- VMs are called **unprivileged domain** (DomU)
- The hypervisor replaces the host operating system
 - But the developers cannot develop all drivers from scratch and maintain them
 - Therefore, the hypervisor launches an (Linux) instance with its drivers and borrows them
 - This instance is called Domain0 (Dom0)

• Drawbacks:

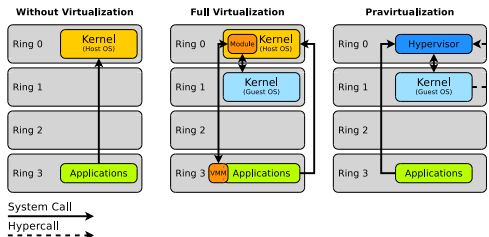
- Kernels of guest operating systems must be modified (adapted) for operation in the paravirtualized context
- Rights holders of proprietary operating systems often reject an adjustment because of strategic reasons
 - ⇒ Often works only with open source operating systems

• Advantage:

- Better performance compared with VMM implementations

Summary: Virtualization vs. Paravirtualization

- **Paravirtualization** requires modified guest systems
 - Type-1 hypervisor runs *bare metal* (= replaces the host operating system)
 - Hypervisor runs in ring 0 and has full access to the hardware
 - Examples: VMware ESX(i), Xen, Microsoft Hyper-V
- **Full virtualization** supports unmodified guest systems
 - VMM (Type-2 hypervisor) runs *hosted* as an application in the host operating system
 - VMM runs in ring 3 at the level of the applications
 - Examples: VMware Workstation, KVM, Oracle VirtualBox, Parallels

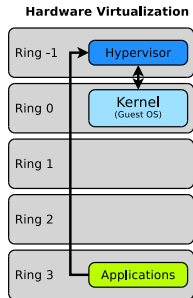


Hardware-assisted Virtualization (1/2)

- Modern CPUs from Intel and AMD contain virtualization extensions for hardware-assisted virtualization
 - Advantage: Unmodified operating systems can be used as guest systems
 - The solutions from Intel and AMD are similar but incompatible
 - Since 2006, AMD64 CPUs contain the Secure Virtual Machine (**SVM**) instruction set
 - The solution is called **AMD-V** and was previously called **Pacifica**
 - The solution from Intel is called **VT-x** for IA32 CPUs and **VT-i** for Itanium CPUs
 - The solution of Intel was previously called **Vanderpool**
- Since Xen version 3, the software supports hardware virtualization
 - Windows Server 2008 (Hyper-V) uses hardware virtualization
 - VirtualBox supports hardware virtualization
 - KVM can only operate with CPUs, which implement hardware virtualization

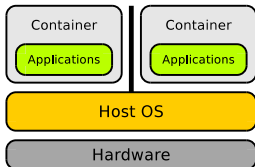
Hardware-assisted Virtualization (2/2)

- The hardware virtualization implementation contains a modification of the privilege levels
- A new ring ($\Rightarrow$ ring -1, called *root mode*) for the hypervisor is added
 - The hypervisor or VMM runs in ring -1 and at any time has the full control over the CPU and the other resources, because with ring -1 an increased privilege level is implemented compared with ring 0
- VMs executed inside ring 0 are called HVM
 - HVM = Hardware Virtual Machine
- Advantages:
 - Guest operating systems do not need to be modified (adapted)
 - Even proprietary operating systems (e.g. Windows) can be used as guest systems



Operating System-level Virtualization / Containers (1/2)

- Under a single kernel, multiple identical, isolated system environments are executed
 - No additional operating system is started
 - Isolated runtime environments are created
 - All running applications use the same kernel
 - This kind of virtualization is called **Containers** in SUN/Oracle Solaris
 - This kind of virtualization is called **Jails** in BSD

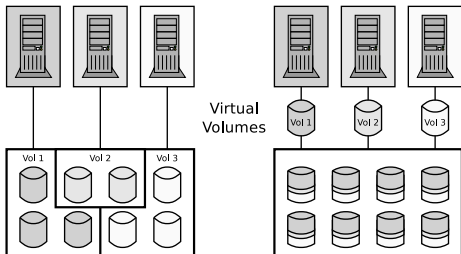


- Applications only see applications from the same virtual environment
- One advantage is the low overhead, because the kernel manages the hardware as usual
- Drawback: All virtual environments use the same kernel
 - Only independent instances of the same operating system are started
 - It is impossible to start different operating systems at the same time

Operating System-level Virtualization / Containers (2/2)

- This type of virtualization is used to execute applications in isolated environments with high security
- Especially internet service providers, which offer (virtual) root servers, or web services on multi-core processor architectures, use this type of virtualization
 - Little performance loss, high security level
- Examples:
 - SUN/Oracle Solaris (2005)
 - OpenVZ for Linux (2005)
 - Linux-VServer (2001)
 - FreeBSD Jails (1998)
 - Parallels Virtuozzo (2001, commercial version of OpenVZ)
 - FreeVPS
 - Docker (2013)
 - chroot (1982)

Storage Virtualization



- Advantages:

- Users are independent from the physical limits of drives
- Reorganizing/expanding the physical storage does not disturb the users
- Redundancy is provided transparently in the background
- Better degree of utilization, because the physical storage can be split among the users in a more efficient way

- Drawback: Professional solutions are expensive

- Some Providers: EMC, HP, IBM, LSI and SUN/Oracle

- Storage is provided to users in form of virtual drives (*volumes*)
- Logical storage is separated from physical storage

Network Virtualization via Virtual Local Area Networks

- Distributed devices can be combined via VLAN in a single virtual (logical) network
 - VLANs separate physical networks into logical subnets (overlay networks)
 - VLAN-capable switches do not forward packets of a VLAN into other VLANs
 - A VLAN is a network, over existing networks, which is isolated from the outside
 - Devices and services, which belong together, can be consolidated in separate VLANs
 - Advantage: Other networks are not influenced
⇒ Better security level

Helpful sources

Benjamin Benz, Lars Reimann. *Netze schützen mit VLANs*. 11.9.2006

<http://www.heise.de/netze/artikel/VLAN-Virtuelles-LAN-221621.html>

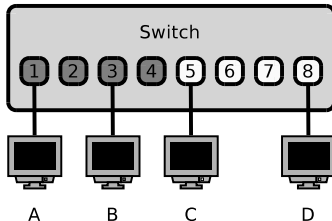
Stephan Mayer, Ernst Ahlers. *Netzsegmentierung per VLAN*. c't 24/2010. S.176-179

100

1 Oldest Standard: **Static VLAN**

- The ports of a switch are assigned to logical switches
- Each port is permanently assigned to a VLAN or it connects different VLANs
- Difficult to automate

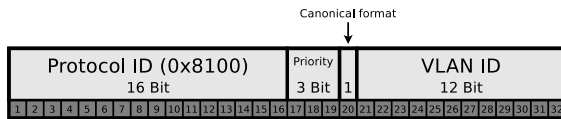
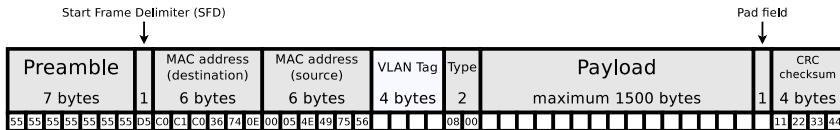
Only node A and B, as well as node C and D can communicate with each other, even though they are all connected with the same Switch



2 Latest: **Packet-based, dynamic VLAN** according to IEEE 802.1Q

- Network layer packets contain a special VLAN *tag*
- Dynamic VLANs can be created, changed and removed purely via software, using scripts

Ethernet Frame with VLAN Tag according to IEEE 802.1Q



- VLAN tag length: 32 bits

Structure of the VLAN Tag

- Protocol ID (16 bits) is always set to 0x8100 for IPv4
- 3 bits are used to store the priority (QoS) value. With this priority value, certain data (e.g. VoIP) can be prioritized
- Canonical format (1 bit) $\Rightarrow$ most significant bit of the MAC addresses
0 = Ethernet, 1 = Token Ring
- 12 bits contain the ID of the VLAN to which the packet in the frame belongs

Examples of Useful Application Areas for VLANs

● Telekom Entertain

- DSL connection with telephone line and IPTV ($\implies$ *Triple Play*)
- Uses 2 VLANs to transmit the IPTV traffic with a higher priority
 - „Normal“ internet via PPPoE via VLAN ID 7
 - IPTV without dialing via VLAN ID 8

● Eucalyptus (<https://www.eucalyptus.cloud>)

- Private cloud infrastructure service (IaaS)
- Each virtual machine (instance) is assigned to a security group
 - Each security group has its own firewall rules
- Eucalyptus can create a separate VLAN for each security group
 - Isolation of the traffic of instances according to the security groups

● Data centers or home office

- Separation of the traffic according to economic aspects
- Objective: Protect against operator errors and defective software
 - One VLAN for a „production network“ with the critical services
 - Additional VLANs for experiments, project work or children's games

Reasons for using Virtualization

- Better hardware utilization
 - Server consolidation: Merge (virtual) servers on fewer physical servers
 - Less cost for hardware, electric energy, cooling, floor space, etc.
- Simplified administration
 - Number of physical servers is reduced
 - Sophisticated management tools exist
 - VMs can be relocated, duplicated and backed up during operation
 - Snapshots of the current state of a VM can be created and restored
- Simplified deployment
 - Infrastructures and servers can be started (automatically!) within minutes
- Increased security level
 - VMs are isolated from other VMs and the host system
 - Critical applications can be encapsulated and run in a secure environment
 - Failure of a VM does not influence other VMs or the host
- Support for legacy operating systems or legacy applications
 - Independence from obsolete hardware (End-of-Life-Hardware)